Computer assisted discovery and open coding

Digital methods lecture 9

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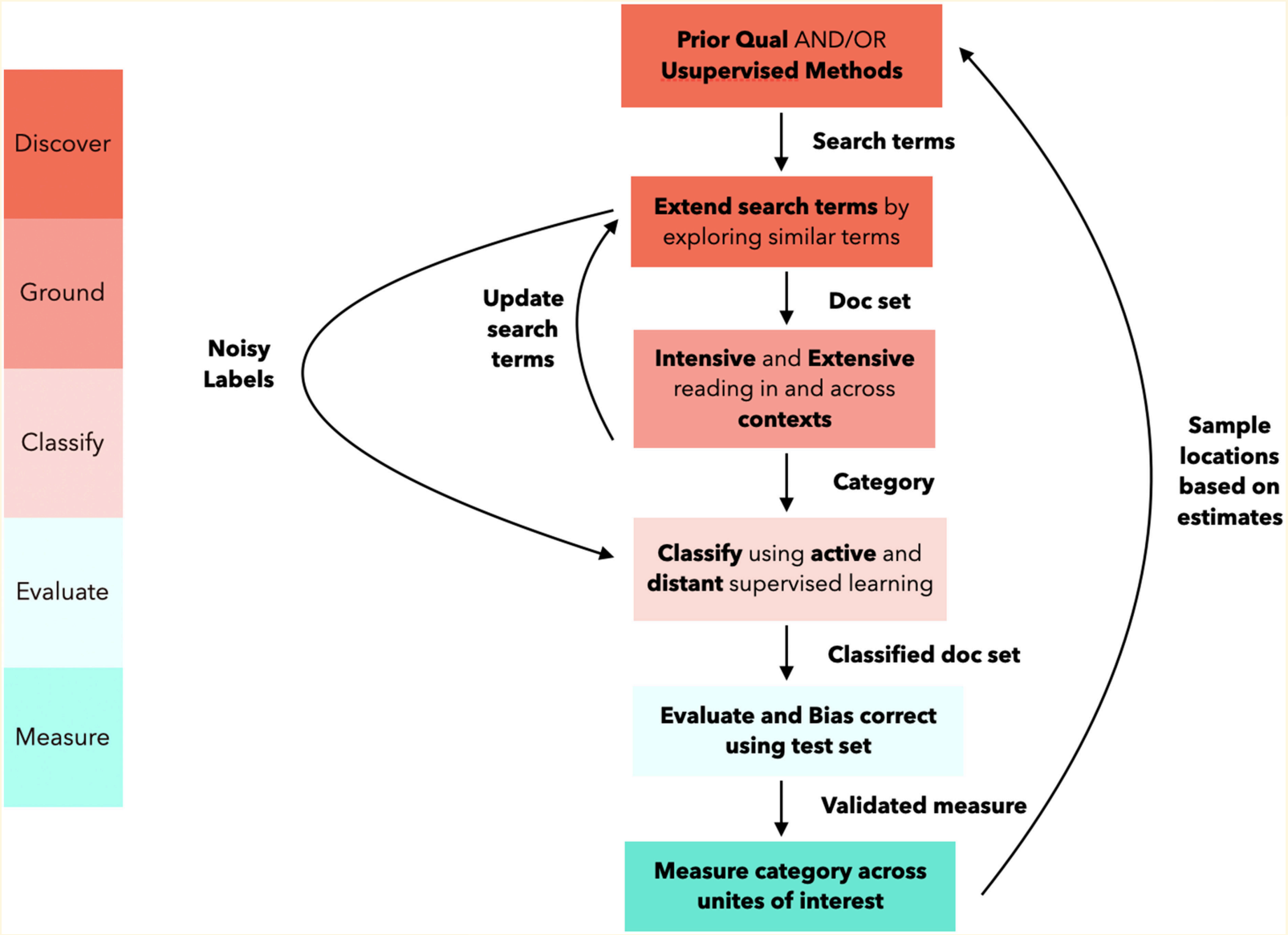
Todays task

1. Introduce the computer assisted framework for content analysis
2. Computer assisted discovery of text categories
3. Open coding
4. In class open coding session.
5. In class illustration of the workflow

Introduce the computer assisted framework for content analysis

From computer lead to computer assisted text analysis

1. Discovery: **from determining** the universe of categories to **helping** with their **discovery**
2. Refinement/grounding: from relying on top documents to extensive and intensive reading using machine learning methods to effectively sample documents
3. Classification, validation and measurement: from indirect to direct validation, from unsupervised to supervised classification.


Discover
Ground
Classify
Evaluate
Measure
Prior Qual AND/OR Usupervised Methods
Search terms
Extend search terms by exploring similar terms
Doc set
Intensive and Extensive reading in and across contexts
Category
Classify using active and distant supervised learning
Classified doc set
Evaluate and Bias correct using test set
Validated measure
Measure category across unites of interest
Noisy Labels
Update search terms
Sample locations based on estimates

Computer assisted discovery of text categories

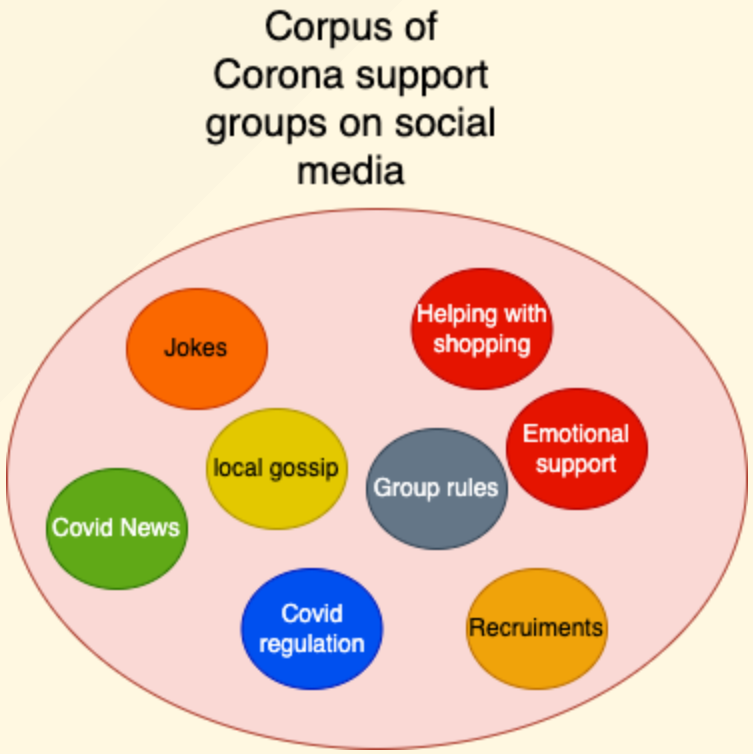
"…can surprise, challenging presumptions or pre-existing theory, and lead the social analyst to abductively generate new theory by imagining what would be socially required for those patterns to exist" (Evans and Aceves, 2016, 23).

Computer assisted discovery of text categories

1. Separate the phase of discovery from that of grounding and measurement
2. Take unsupervised methods, words counts(aso.) as highly uncertain proxies for categories
3. Yet, useful initial mapping, navigation, sampling and theorizing
4. To support, but not determine the researchers exploration

What can we use unsupervised methods for?

1. Uncertain sense of the **discursive context**

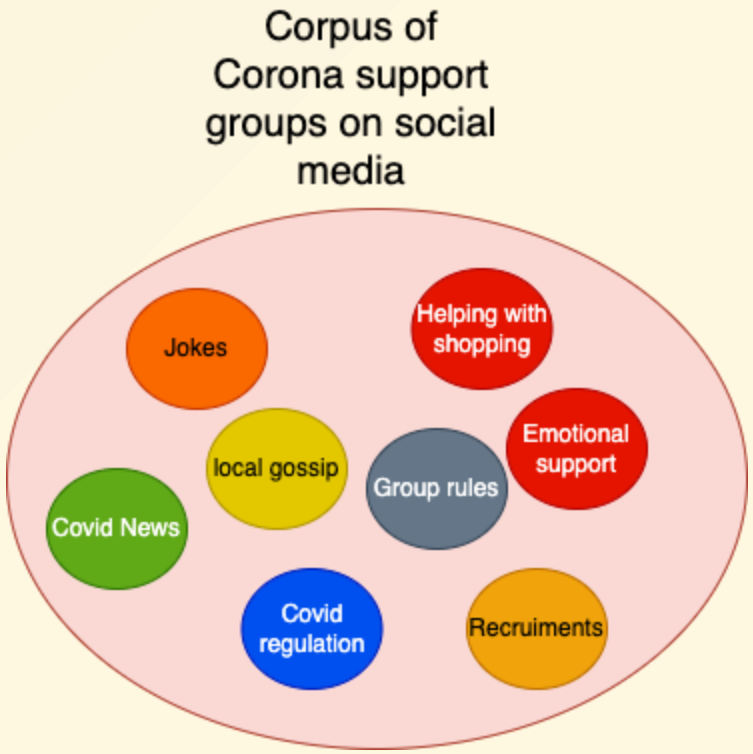


What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
2. **Defamiliarization**: see your datasite from a unfamiliar perspectives which provokes you to rethink how you conceptualize your object of study

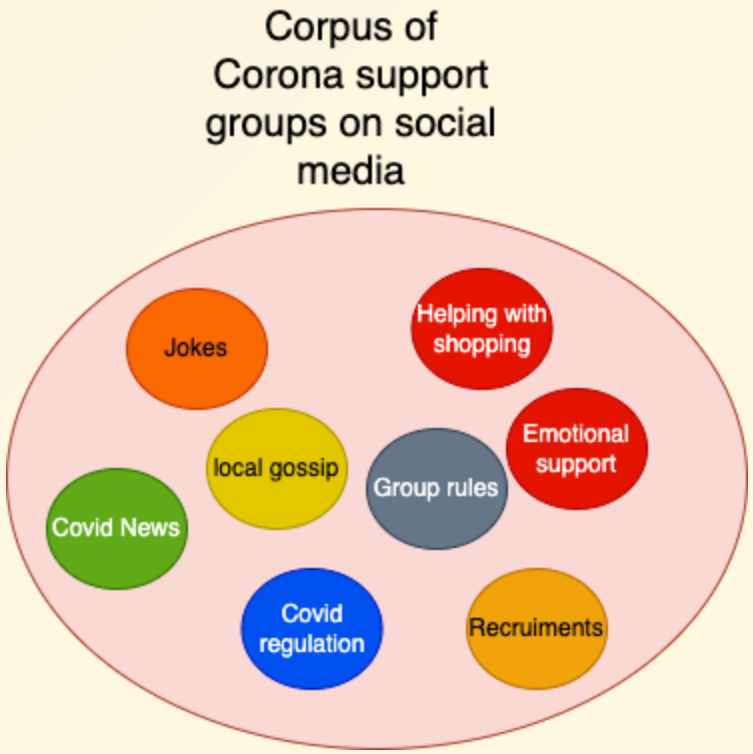
What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
2. **Defamiliarization**



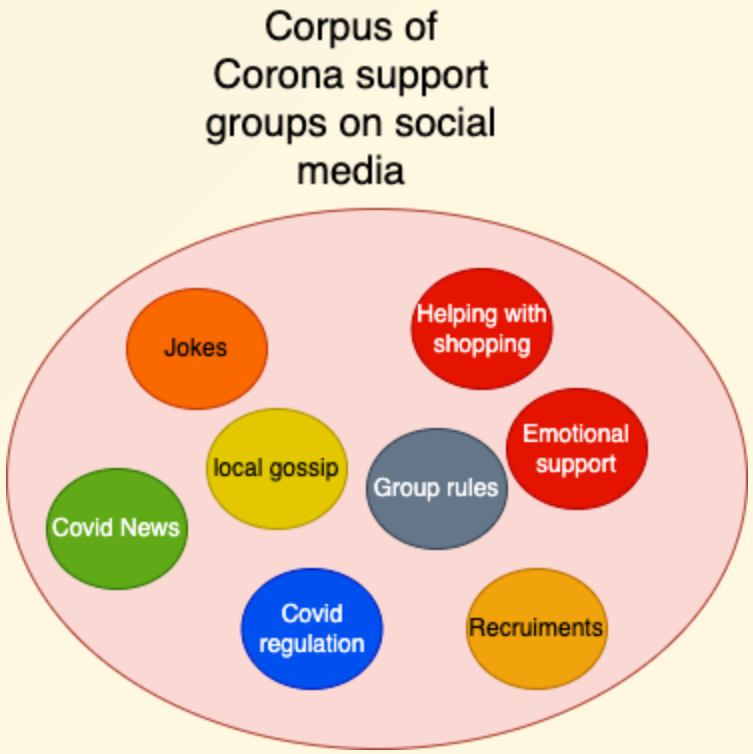
What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
2. Defamiliarization
3. **Propose new categories**



What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
2. Defamiliarization
3. Propose new categories
4. Propose a **split** of category



What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
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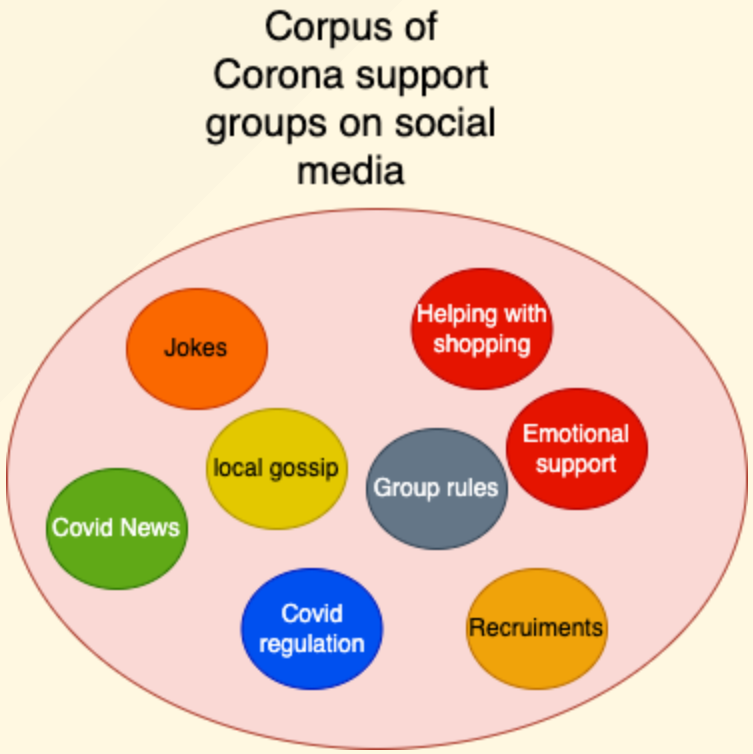
3. Propose new categories
4. Propose a split of category
5. Propose **lumping** two categories together



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1. Uncertain sense of the discursive context
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6. Provide **keywords** that we use to **sample documents** AND as **seeds**

What can we use unsupervised methods for?

1. Uncertain sense of the discursive context
2. Defamiliarization
3. Propose new categories
4. Propose a split of category
5. Propose lumping two categories together
6. Provide keywords
7. **Explore** the discursive context of **central category**

What methods?

1. LDA, Structural Topic Model
2. HSBM hierarchical stochastic blockmodels
3. Semantic Network Analysis
4. Different types of word counts
5. And more

Unsupervised methods for keyword discovery - why?

1. Use keywords to sample documents relevant to understanding a certain category/aspect of your datasite.
2. Use extensive keyword lists to capture as many relevant documents as possible - to ensure exposure
3. Use extensive keyword lists understand and explore the full variation of ones categories
4. Use extensive keyword lists for downstream classification

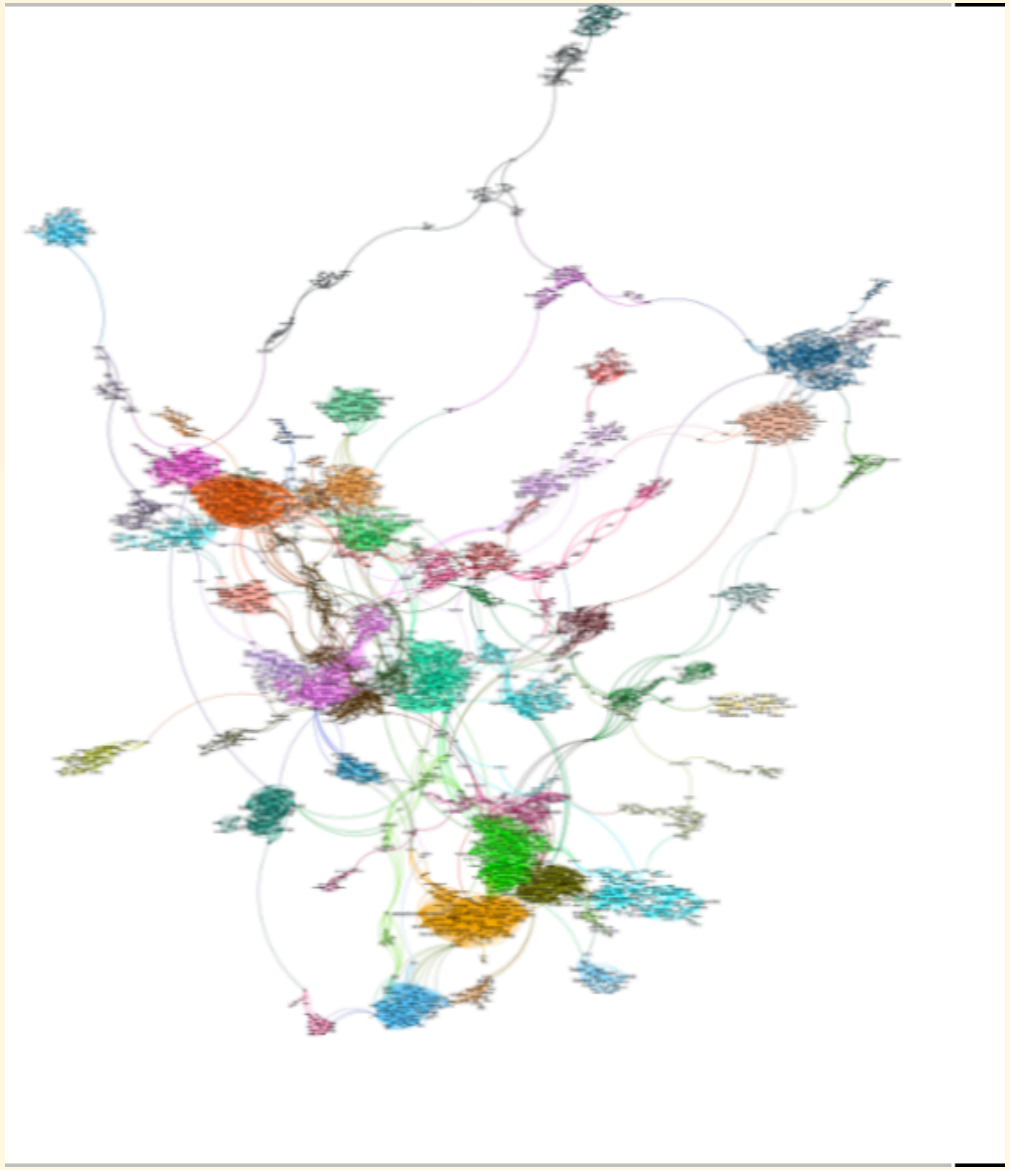
Unsupervised methods for keyword discovery - how?

1. Run the model
2. **Inspect clusters of words**
    - **LDA top words within a topic**

mujahideen, kill, attack, soldier, captur, enemi, oper, command, tank, area
pepe, pace, hair, crush, bottom, fashion, mesa, dam, code, matt
god, jihad, quot, messeng, allah, brother, prophet, sheikh, bless, martyr
educ, thousand, provinc, news, train, respect, ministry, school, abdul, mullah
mujahedeen, jihad, soviet, najib, kabul, regim, afghan, elect, presid, pakistan
movement, holy, life, great, human, scholar, women, quran, answer, world
afghanistan, russian, countri, govern, unit, intern, russia, support, state, american
social, uniti, polit, communiti, revolut, parti, west, exist, natur, system
grant, sunday, catalog, prison, friday, inevit, servic, pose, pro, innov
profil, breath, select, patent, post, bbc, threat, bar, tuesday, cost

Unsupervised methods for keyword discovery

1. Run the model
2. Inspect clusters of words
   - LDA top words within a topic
   - **Network: explore clusters**



Unsupervised methods for keyword discovery

1. Run the model
2. Inspect clusters of words
3. **Build a category and connected keyword/seed lists**

Category	Seed words from Netnography	Seed words from unsupervised ML
cate 1	word 1, word 2, word 3	word 2, word 3, word 99
cate 2	word 1, word 2	word 5, word 10, word 19
cate 3		word 4, word 5

Methods for keyword expansion

1. Computer assisted keyword discovery from King et al. 2017

Which non-seed words are predictive of the documents that my seeds words are in?

Methods for keyword expansion

1. Computer assisted keyword discovery from King et al. 2017
2. **Word embeddings, word2vec**

Which words share the same context as my seed words

Methods for keyword expansion

1. Computer assisted keyword discovery from King et al. 2017
2. Word embeddings, word2vec
3. **word2vec can take both lists and single words as input**

Methods for keyword expansion

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4. **Populate your keyword lists**

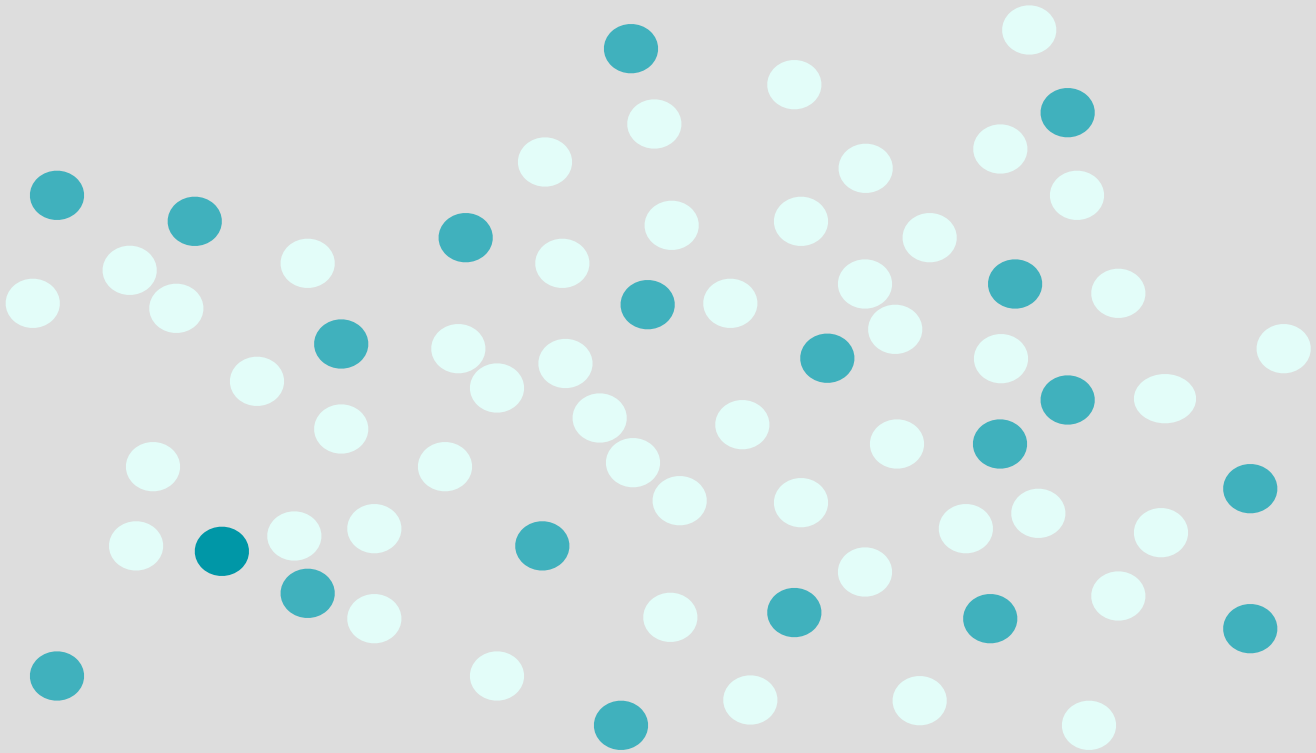
Methods for keyword expansion

1. Computer assisted keyword discovery from King et al. 2017
2. Word embeddings, word2vec
3. word2vec can take both lists and single words as input
4. Populate your keyword lists
5. **Sample documents using keywords as query terms**

Seed-words from Network	keywords after w2v extension
'brune' 'diskrimination', 'hudfarve', 'racisme', 'racist', 'racistisk', 'racistiske', 'etniske'	islamofobi racebevidsthed farveblindhed udlændingelov nationalisme nazisme neger racist race forskelsbehandling n-ordet raceliseret ikke-hvide sorte_mennesker minoriteter etniske racisme stereotyper racer diskrimination racialiserede fordomme hudfarve racister undertrykkelse hvide_mennesker strukturel_racisme nazister krænkende

Keyword expansion as snowball sampling

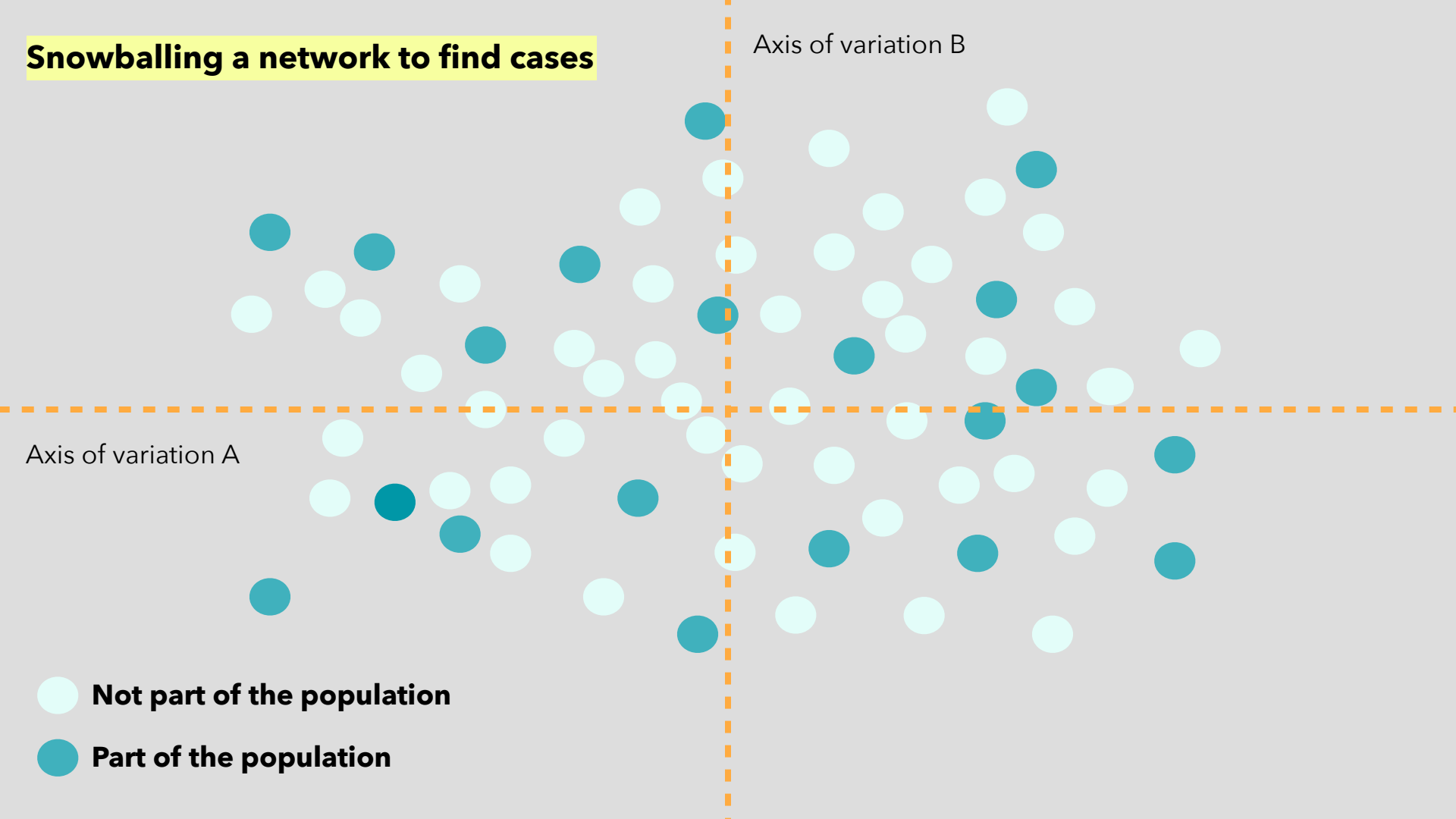
Snowballing a network to find cases

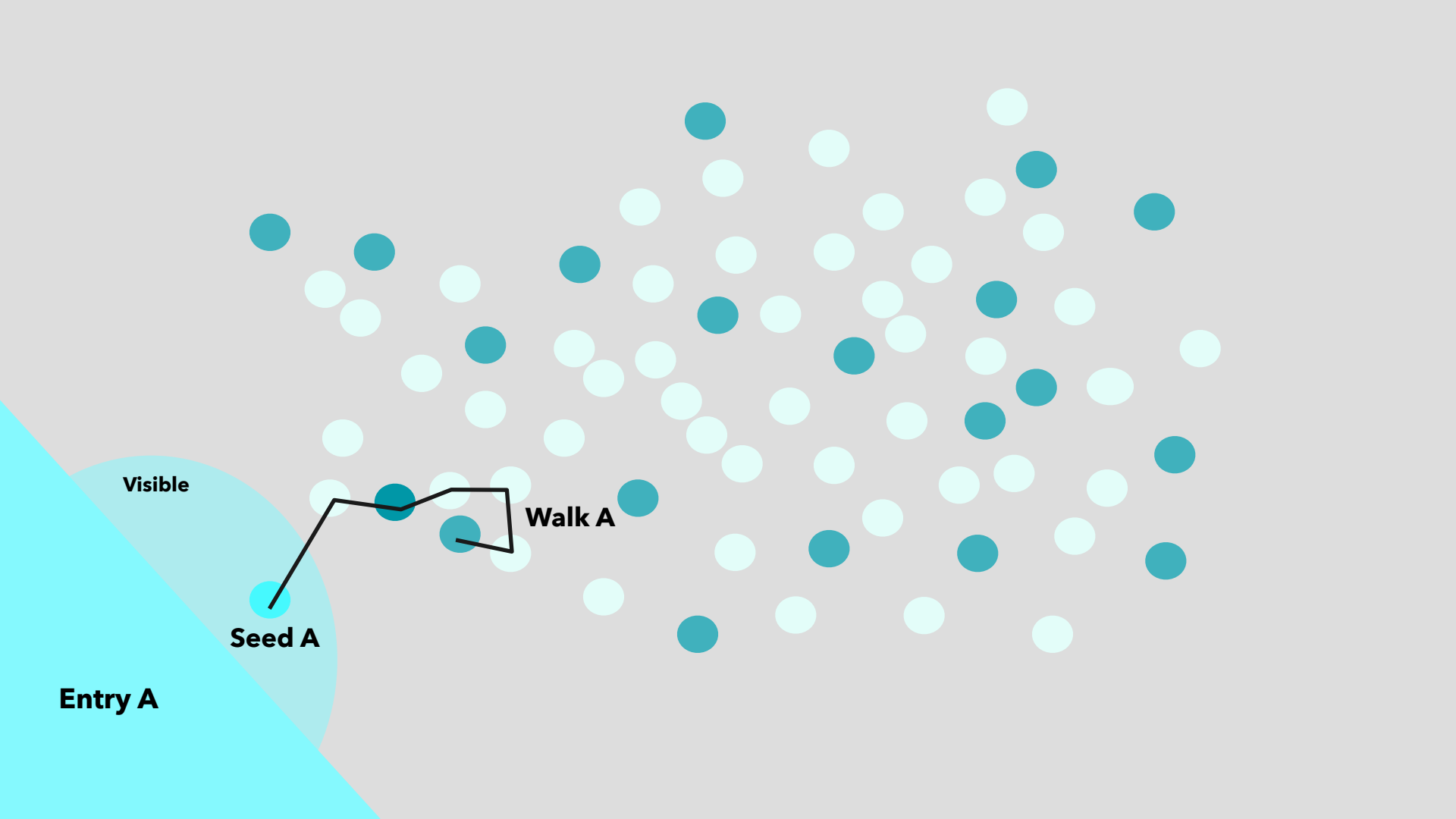


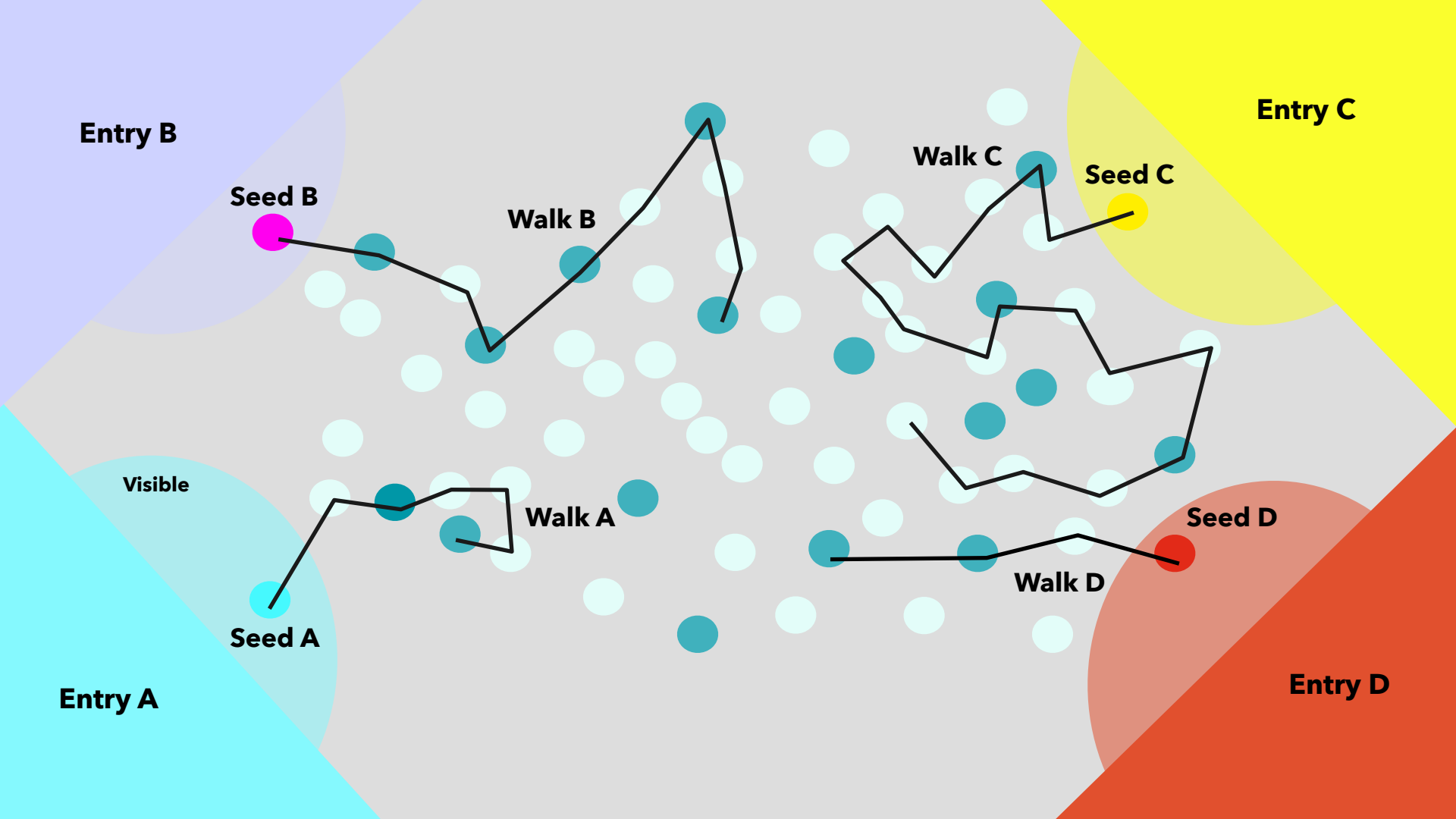
Not part of the population

Part of the population

Snowballing a network to find cases

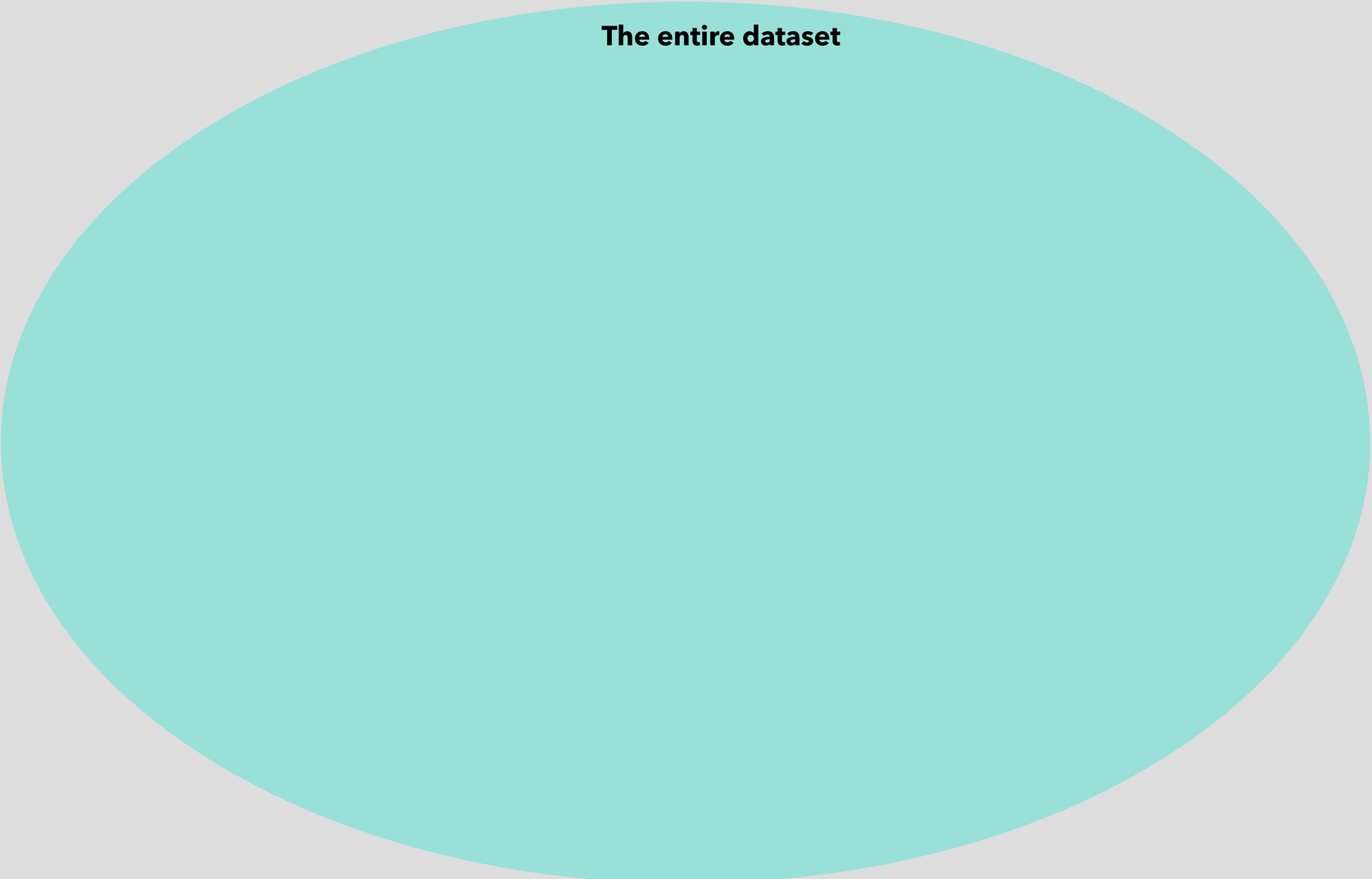



Visible
Walk A
Seed A
Entry A

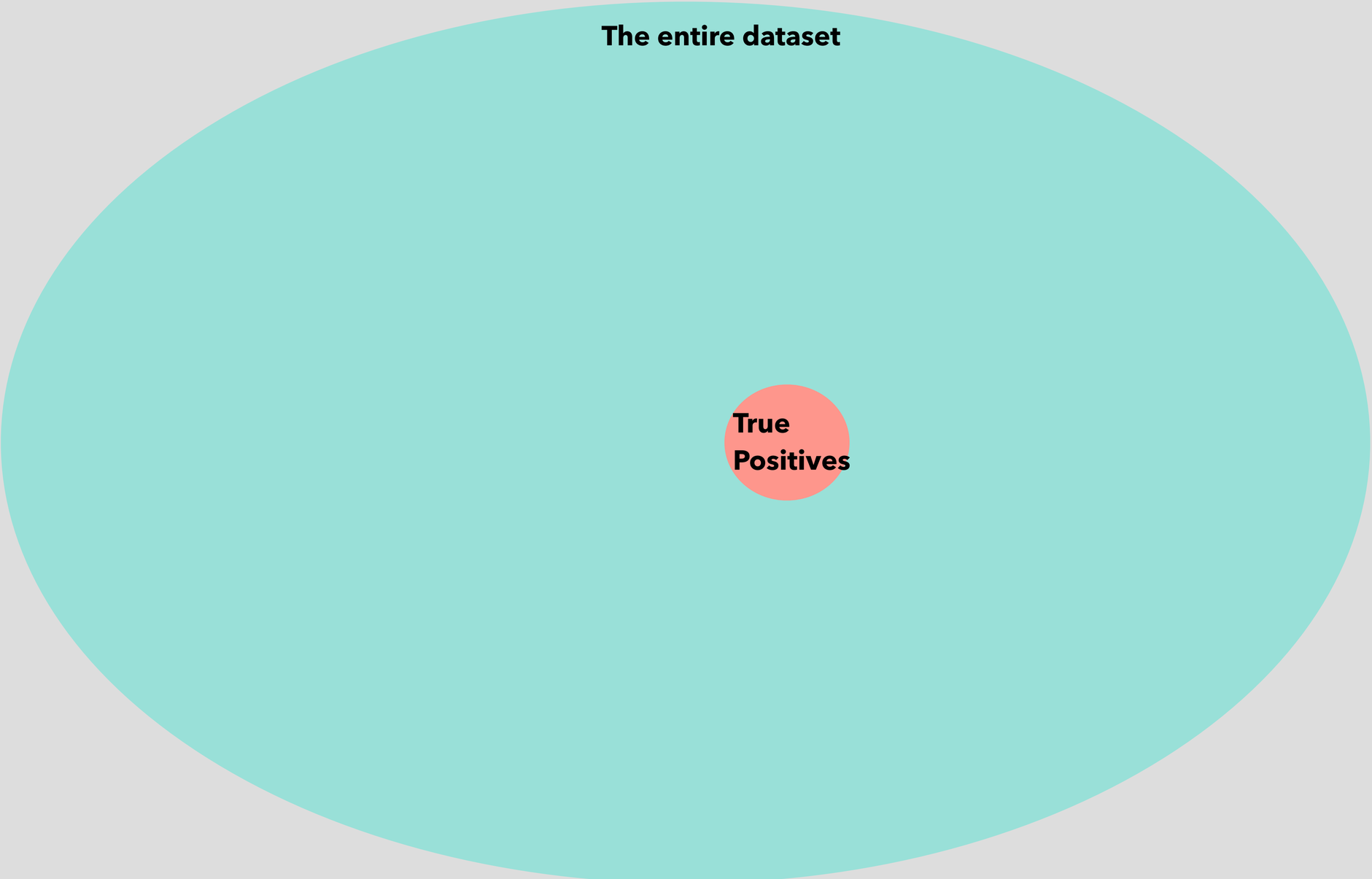

Entry B
Entry C
Seed B
Walk B
Walk C
Seed C
Visible
Walk A
Seed D
Walk D
Seed A
Entry A
Entry D

Keyword expansion for document sampling

Precision Recall dilemma



Precision Recall dilemma



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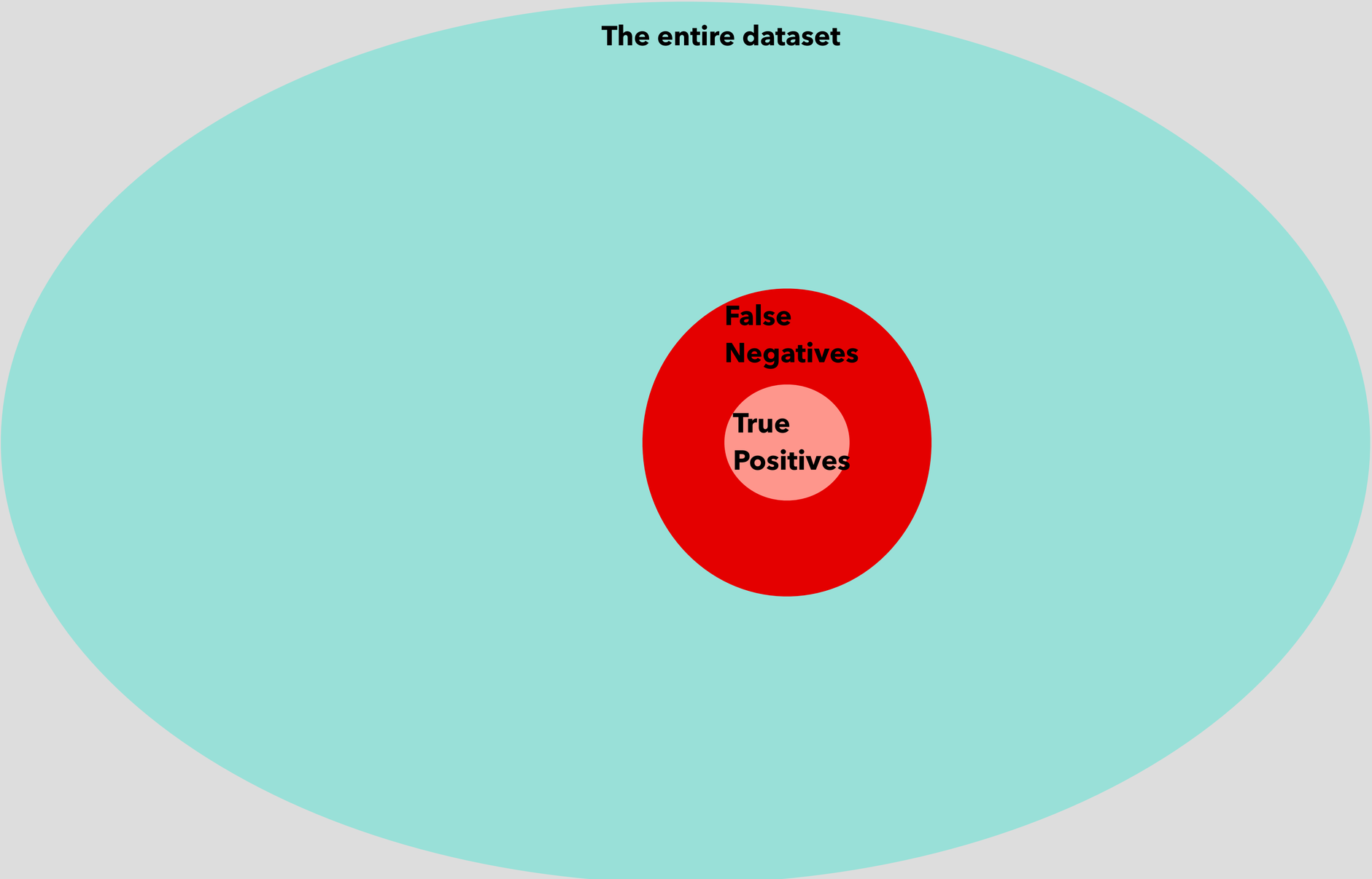

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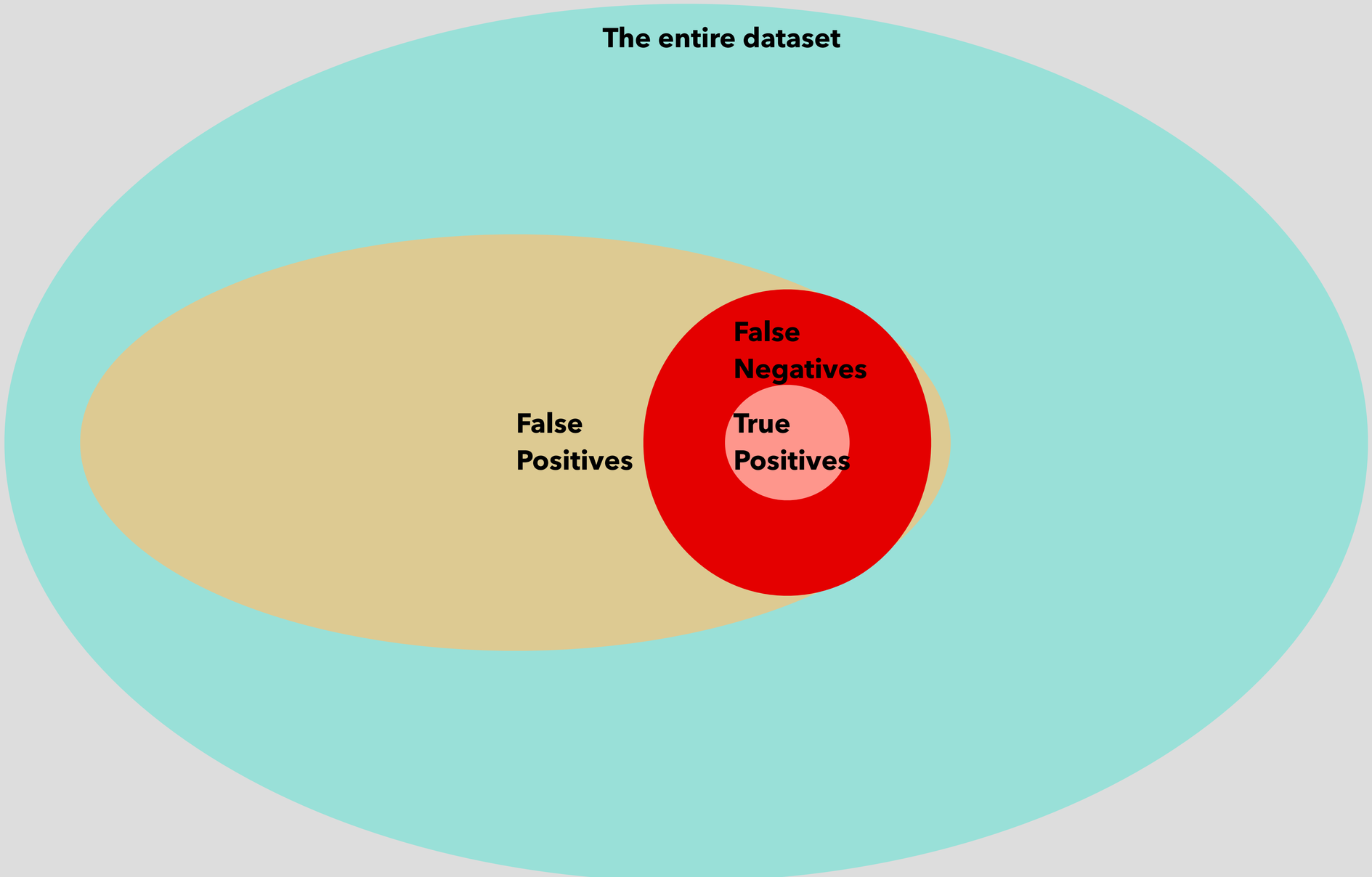

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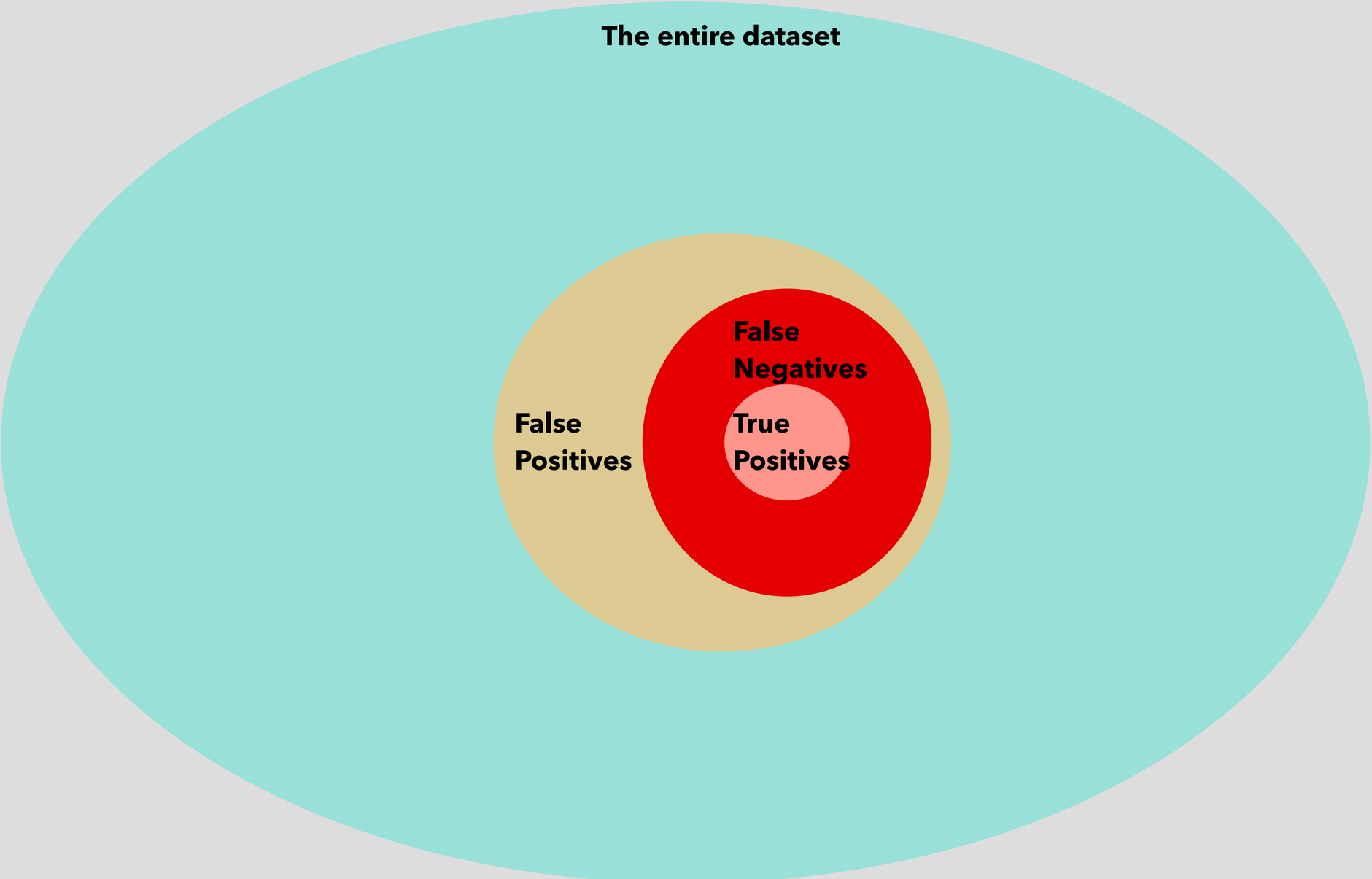

What information do then need to interpret your data?

- Full thread?
- Author information?
- Temporal information?
- Social media feedback(likes aso)

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- Full thread?
- Author information?
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Think about what information you need in order to most efficiently interpret the content in a sound manner.

Open Coding

Open Coding - why?

1. To gain extensive understanding of datasite
2. Validate or alter our initial understanding of a category
3. Locate interesting theoretical perspectives
4. Find data elements that can be compared for their analytical properties
5. Locate analytical interests(for focused coding)

Open Coding - what?

1. A code moves from the concrete observation to some abstract category

What does a code do? It labels a snippet of an observation as a particular "kind." Then, by putting other data instances under the conceptual umbrella of the code, you develop what this kind is about. With coding, you abstract away from observation to categorization. You interrogate the minute differences between observations, while temporarily suspending these differences by

Open Coding - what?

1. A code moves from the concrete observation to some abstract category
2. It forces you to answer "what is this a case of"

Open Coding - how?

1. We use various heuristics to start your investigation
2. **Who did what, when, where, how and with what consequences**

Open Coding - how?

1. We use various heuristics to start your investigation
2. **Who did what, when, where, how and with what consequences**

Agency, purpose, temporal sequence, social setting, social mechanism, consequences

Open Coding - how?

1. We use various heuristics to start your investigation
2. Who did what, when, where, how and with what consequences
3. **Think/reread Abbott heuristics to inspire reading of your material**

Reconceptualize, changing levels, Reversals

Open Coding - how?

1. We use various heuristics to start your investigation
2. Who did what, when, where, how and with what consequences
3. Think/reread Abbott heuristics to inspire reading of your material
4. Actively work with the theories relevant to your study
5. **A code, a short memo/analysis**

Open Coding collectively!

Read, analyze and code individually or in groups focusing on:

1. Try to answer the initial "Who did what, when, where, how and with what consequences"
2. 2-3 Interesting theoretical perspectives given the focus of the study
3. 1-2 Alternative theoretical perspectives prominent in the data
4. Note the relevant place in the text, provide an initial code name

We have three examples with 20 min per example. 10 min for analyzing and coding. 10 min for discussing.

1. case: Political communication from farmers

2. case: Astrology online forum

3. case: Depression subreddit.